YEAR 12 GENERAL MATHEMATICS INVESTIGATIVE TASK 1 SOLUTIONS

Financial applications

Question 1

Ava gets $2000 for her 18th birthday and wants to deposit it into a high-interest account while she thinks about how to spend it. Her current bank offers two options:

- Simple Savings account: 9.75% p.a.
- Clever Compounds account: 7.99% p.a. (compounding yearly).

a How much interest would Ava earn in a year with each account?

Simple Savings: $2000 \times 0.0975 = \$195$

Clever Compounds: $2000 \times 0.0799 = \$159.80$

2 marks

b If Ava left the money in the account for 3 years, what would be the balance of each bank account?

Simple Savings: $2000 + 3 \times 195 = \$2585$

Clever Compounds: $2000 \times 1.0799^3 = \$2518.72$

2 marks

c Complete the table below to determine how many years it would take for the Clever Compounds account to become more beneficial than the Simple Savings account.

Year	Simple Savings account ($)	Clever Compounds account ($)
0	2000	2000
1	$2000 + 195 = 2195$	$2000 \times 1.0799 = 2159.80$
2	$2195 + 195 = 2390$	$2159.80 \times 1.0799 = 2332.37$
3	$2390 + 195 = 2585$	$2332.37 \times 1.0799 = 2518.72$
4	$2585 + 195 = 2780$	$2518.72 \times 1.0799 = 2719.97$
5	$2780 + 195 = 2975$	$2719.97 \times 1.0799 = 2937.30$
6	$2975 + 195 = 3170$	$2937.30 \times 1.0799 = 3171.99$

After 6 years the Clever Compounds account would become more beneficial than the Simple Savings account.

3 marks

Another bank offers a compound interest account, Monthly Savings, earning 7.5% compounding monthly.

d How much interest would Ava earn in the first year if she chose this account?

$$A = 2000\left(1 + \frac{7.5}{1200}\right)^{12} = \$2155.27$$

Interest earned $= \$2155.27 - \$2000 = \$155.27$.

2 marks

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Ava's friend, Isaac, tells her about the savings account, Savings Quarterly, that he has been using for the past 3 years. He can't remember the interest rate, but he knows that he started with a deposit of $10 000, and after 3 years of compounding quarterly he now has $12 534.01.

e Find the interest rate of Isaac's savings account.

$$A = P\left(1 + \frac{r}{100}\right)^n$$

$$12534.01 = 10000\left(1 + \frac{\frac{r}{4}}{100}\right)^{12}$$

$$1.253401 = \left(1 + \frac{\frac{r}{4}}{100}\right)^{12}$$

$$1.019 = 1 + \frac{\frac{r}{4}}{100}$$

$$0.019 = \frac{\frac{r}{4}}{100}$$

$$\frac{r}{4} = 1.9$$

$$r = 7.6$$

Isaac's interest rate is 7.6%, compounding quarterly.

2 marks

f If she chose this account, what would be Ava's account balance at the end of 3 years?

$$A = P\left(1 + \frac{r}{100}\right)^n$$

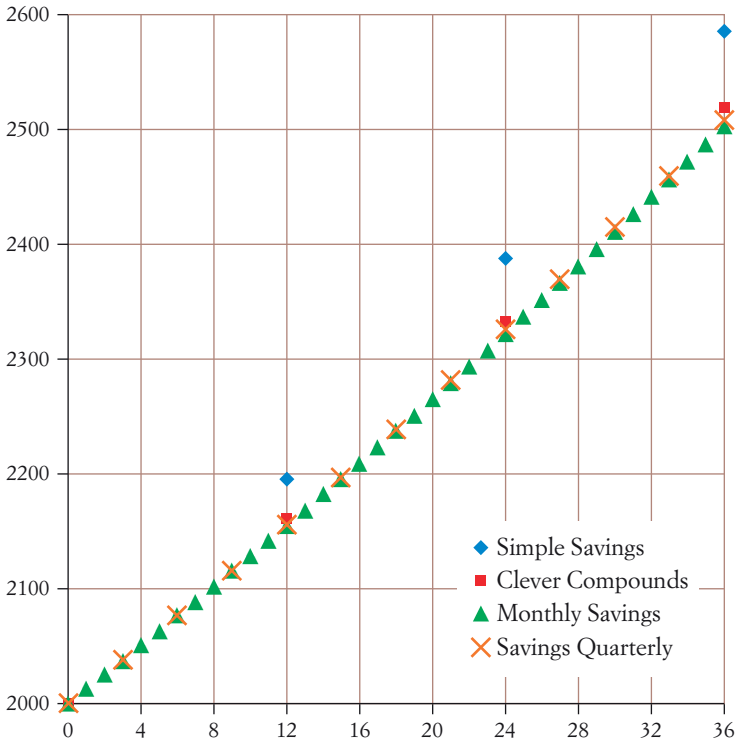
$$A = 2000\left(1 + \frac{\frac{7.6}{4}}{100}\right)^{12} = 2506.80$$

Ava's account balance will be $2506.80 at the end of 3 years.

1 mark

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g Draw a single graph showing the growing balance for these four savings accounts over 3 years.



3 marks

h If Ava wanted to deposit her money for 10 years, which account would you recommend that she choose?

Simple Savings: $A = 2000 + 195 \times 10 = 3950$

Clever Compounds: $A = 2000\left(1 + \frac{7.99}{100}\right)^{10} = 4313.85$

Monthly Savings: $A = 2000\left(1 + \frac{\frac{7.5}{12}}{100}\right)^{120} = 4224.13$

Savings Quarterly: $A = 2000\left(1 + \frac{\frac{7.6}{4}}{100}\right)^{40} = 4246.17$

The best option for Ava over a 10-year period is the Clever Compounds account.

3 marks

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Question 2

Finn goes shopping for a new laptop. He sees one he wants for $1699 at Lappys, but cannot afford that much up front. The store offers him a deal where he pays a 10% deposit, and then $79 a month for 2 years.

a How much interest will Finn pay if he accepts this deal?

Finn pays $\$169.90 + \$79 \times 24 = \$2065.90$

Interest paid $= \$2065.90 - \$1699 = \$366.90$

2 marks

b What is the flat rate of interest charged per annum on the laptop?

Loan $= \$1699 - \$169.90 = \$1529.10$

$$I = \frac{Prn}{100}$$

$$366.90 = \frac{1529.10 \times r \times 2}{100}$$

$$I = \frac{Prn}{100}$$

$$366.90 = \frac{1529.10 \times r \times 2}{100}$$

$$r = \frac{366.90 \times 100}{1529.10 \times 2}$$

$$= 11.9973$$

The interest rate is approximately 12% per annum.

2 marks

The competing store, Technica, offers the same laptop for $1850 under a 'Buy now, pay later' deal. Finn was able to take home the laptop that day and pay nothing for 14 months. Any outstanding balance will be charged at an interest rate of 5% per month from the date of purchase.

c How much will Finn need to pay per month towards this laptop if he is to pay it off before the 14 months are up?

$\$1850 \div 14 = \$132.14$

1 mark

d If Finn can pay the full $1850 before the 14 months are up, should he take this deal or the one from Lappys?

Yes, he should take this deal, as $1850 is cheaper than $2065.90.

1 mark

e Finn is able to put $110 per month towards the laptop. How much will he still owe just before the end of the 14 months?

$\$1850 - \$110 \times 14 = \$310$

1 mark

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f Calculate the interest that will be charged on the remaining balance and, hence, state what Finn will now owe when the 14 months are up.

$\$310 \times 1.05^{14} = \$613.78$

1 mark

g Copy and complete the table below to show how long it will take Finn to pay back the loan if he continues to put \$110 a month towards the laptop. Assume that Finn pays \$100 at the start of the month and that the interest is calculated at the end of the month.

Month	Balance (\$)	Minus payment (\$)	Balance after payment (\$)	Plus interest (\$)	End balance (\$)
14	613.78	110	503.78	25.19	528.97
15	528.97	110	418.97	20.95	439.92
16	439.92	110	329.92	16.50	346.41
17	346.41	110	236.41	11.82	248.23
18	248.23	110	138.23	6.91	145.14
19	145.14	110	35.14	1.76	36.90
20	36.90	36.90	0		

Finn will pay off the loan at the start of the 20th month.

4 marks

h Use the table from part **g** to show which deal Finn should take.

Total paid on original offer is \$2065.

Total paid on second offer is $110 \times 19 + 36.90 = \$2126.90$.

If Finn is unable to keep up with payments, he is better to take the original offer.

2 marks

Question 3

Olivia purchases a new house for \$575 000. She has \$80 000 for the deposit and must borrow the remaining balance from the bank. Her bank offers her a 30-year loan with an interest rate of 6.75% compounding monthly.

a What will be Olivia's monthly repayments?

$PV = \$495\,000$

$I = 6.75$

$n = 30 \times 12 = 360$

$FV = 0$

$PMT = -3210.56$

Payment = \$3210.56 per month.

1 mark

b How much interest will Olivia pay over the life of the loan?

$\$3210.56 \times 360 - \$495\,000 = \$660\,802$

2 marks

After 10 years, Olivia is able to raise her monthly repayments to \$4000.

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c How many months will Olivia save on her loan if she pays $4000 a month for the remainder of the loan?

First 10 years:

$PV = 495\,000$

$I = 6.75$

$n = 10 \times 12 = 120$

$PMT = -3210.56$

$FV = \$422\,239.95$

Remainder of the loan:

$PV = 422\,239.95$

$I = 6.75$

$PMT = -4000$

$FV = 0$

$n = 161$ months

360 months $-$ (120 $+$ 161) $=$ 79 months

Olivia will save 79 months, or 6 years and 7 months.

3 marks

d How much interest will she save?

Original interest to be paid $=$ $660 802.

Olivia pays $\$3210.56 \times 120 + \$4000 \times 161 = \$534\,267$.

Olivia saves $\$660\,802 - \$534\,267 = \$126\,535$ in interest.

2 marks

After a further 5 years, the bank raises the interest rate to 7.39%.

e Calculate how much interest Olivia will now pay over the life of the loan.

Amount owing after 15 years:

$N = 60$

$PV = 422\,239.95$

$I = 6.75$

$PMT = -4000$

$FV = 306\,658.29$

Balance at the end of 15 years $=$ $306 658.29.

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New length of loan accounting for change in interest rate:

$PV = 306\ 658.29$

$I = 7.39$

$PMT = -4000$

$FV = 0$

$N = 104$

Payments made in first 10 years $= 3210.56 \times 120 = 385\ 267$.

Payments made in next 5 years $= 4000 \times 60 = 240\ 000$.

Payment of remainder $= 4000 \times 104 = 416\ 000$.

Total paid $= \$1\ 041\ 267$.

Total interest paid $= \$1\ 041\ 267 - 495\ 000 = \$546\ 267$.

4 marks

After a total of 20 years, Olivia needs to undertake $100 000 worth of renovations on the house, and she redraws on her home loan in order to do so.

f Find her monthly repayments if she chooses to keep the original 30-year term.

Balance at end of 15 years $= \$306\ 658.29$

Balance at end of 20 years:

$N = 60$

$PV = 306\ 658.29$

$I = 7.39$

$PMT = -4000$

$FV = 153\ 954.36$

Balance at end of 20 years $= \$153\ 954.36$

New balance $= \$153\ 954.36 + \$100\ 000 = \$253\ 954.36$.

$PV = 253\ 954.36$

$I = 7.39$

$n = 120$

$FV = 0$

$PMT = -2999.92$

Her monthly repayments would be $2999.92 if she kept the 30-year term.

4 marks

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g If she keeps paying $4000 a month, calculate the:

i total amount of time taken to pay off the loan

$PV = 253\,954.36$

$I = 7.39$

$FV = 0$

$PMT = -2999.92$

$N = 81$

She would spend an additional 81 months.

Total time spent = 20 years + 81 months = 26 years 9 months.

2 marks

ii total amount of interest she pays on the loan.

Payments made in first 10 years = $3210.56 \times 120 = 385\,267$.

Payments made in next 10 years = $4000 \times 120 = 480\,000$.

Payment of remainder = $4000 \times 81 = 324\,000$.

Total paid = $1 189 267.

Total interest paid = $1 189 267 − 495 000 = $694 267.

2 marks

h If Olivia increases her monthly repayments to $10 000, can she limit the interest she pays to $600 000?

$PV = 253\,954.36$

$I = 7.39$

$FV = 0$

$PMT = -10\,000$

$N = 28$

It would take 28 months to pay it off at $10 000 per month.

Payments made in first 10 years = $3210.56 \times 120 = 385\,267$.

Payments made in next 10 years = $4000 \times 120 = 480\,000$.

Payment of remainder = $10\,000 \times 28 = 280\,000$.

Total paid = $1 145 267.

Total interest paid = $1 145 267 − 495 000 = $650 267.

No, at that rate, she will not be able to limit the interest to $600 000.

3 marks

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Question 4

Five years ago, Alex opened an account with an interest rate of 4.95%, compounding monthly. At the beginning of every month he deposited $500.

a Calculate his current account balance.

$N = 60$

$PV = 500$

$I = 4.95$

$PMT = 500$

$FV = -34\ 599.78$

Alex's current account balance is $34 599.78.

1 mark

Alex manages to double his deposit amount for the next 5 years.

b Calculate the account balance at the end of 10 years.

$N = 60$

$PV = 34\ 599.78$

$I = 4.95$

$PMT = 1000$

$FV = -112\ 212.91$

Alex's account balance after 10 years is $112 212.91.

1 mark

c Find how much more interest Alex has earned by doubling his deposit amount.

Alex's account balance after 10 years is $112 212.91.

He deposited $500 \times 60 + 1000 \times 60 = \$90\ 000$ of this, so the interest earned is $\$112\ 212.91 - \$90\ 000 = \$22\ 212.90$.

If Alex kept his monthly deposit at $500:

$N = 120$

$PV = 500$

$I = 4.95$

$PMT = 500$

$FV = -78\ 523.21$

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He would have \$78 523.21 in his account.

He deposited $500 \times 120 = 60\,000$ of this, so the interest earned is \$78 523.21 − \$60 000 = \$18 523.20.

Therefore, he has earned an extra \$22 212.90 − \$18 523.20 = \$3689.71 in interest by doubling his deposit amount.

3 marks

At the end of 10 years, Alex is injured on the job, and is unable to work for 2 years. He is compensated but, after medical bills, he is unable to continue depositing money into his savings account. He returns to work on a part-time basis for the following 3 years, and is able to deposit \$500 a month for those 3 years.

d Find the value of Alex's account 15 years after it was opened.

Alex's account balance after 10 years is \$112 212.91.

Alex's account balance after 12 years:

$N = 24$

$PV = 112\,212.91$

$I = 4.95$

$PMT = 0$

$FV = -123\,865.27$

Alex's account balance after 15 years:

$N = 36$

$PV = 123\,865.27$

$I = 4.95$

$PMT = 500$

$FV = -163\,013.57$

Alex's account balance after 15 years = \$163 013.57.

2 marks

At the end of the 15 years, Alex decides to use the savings as a perpetuity. He switches the money to an investment that pays 5.25% p.a., compounding quarterly.

e How much will he receive each quarter from this perpetuity?

$$Q = \frac{Pr}{100}$$
$$= \frac{163013.57 \times \frac{5.25}{4}}{100}$$
$$= 2139.55$$

Alex will receive \$2139.55 each quarter from this perpetuity.

1 mark

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Alex finds that this isn't quite as much money as he would like.

f If he decides to withdraw \$6000 per quarter, how many years will the money last?

$PV = 163\,013.57$

$I = 5.25$

$PMT = -6000$

$FV = 0$

$PY = 4$

$N = 33.82$

The money will last 34 months, or 8.5 years.

1 mark

g If Alex wants the money to last 10 years, to what amount must he limit his withdrawals every quarter?

$N = 40$

$PV = 163\,013.57$

$I = 5.25$

$PY = 4$

$FV = 0$

$PMT = -5264.38$

Alex would have to limit his withdrawals to \$5264.38 every quarter.

1 mark

h How much longer would the annuity last if, after 5 years, the interest rate increases to 6.9% p.a.? Assume that Alex is withdrawing the amount you calculated in part **g**.

Amount in account after 5 years:

$N = 20$

$PV = 163\,013.57$

$I = 5.25$

$PY = 4$

$PMT = -5264.38$

$FV = -92\,075.12.$

After 5 years, there is still \$92 075.12 in the account.

The remainder:

$PV = 92\,075.12$

$I = 6.15$

$PY = 4$

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$PMT = -5264.38$

$FV = 0$

$N = 21$

It would last a total of $20 + 21$ months $= 41$ months.

This is 1 month longer than the original 40 months (10 years).

3 marks

i Find how much interest Alex will have earned from the account from the time that he opened the account until he drained it.

Alex deposited $500 \times 60 + 1000 \times 60 + 500 \times 36 = \$108\,000$.

He received payments of \$5264.38 for 41 months for a total of \$215 840.

Hence, he earned \$215 840 − \$108 000 = \$107 840 in interest for the duration of the account.

2 marks

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